

DigiT — Client Workspace Technical Specification

Implementable components for the Compliance, Resources & Performance workspaces · v1.0

This document specifies the concrete technical building blocks for DigiT's three client-facing workspaces, organised into a **data & security layer**, an **AI / ML layer**, and an **automation layer**. None of these are standalone builds — they surface through five core engines (RBAC, Universal Search, AI Assistant, Workflow Automation, Analytics) on top of the multi-tenant + Supabase foundation. The architecture and the component-to-module mapping are given at the end.

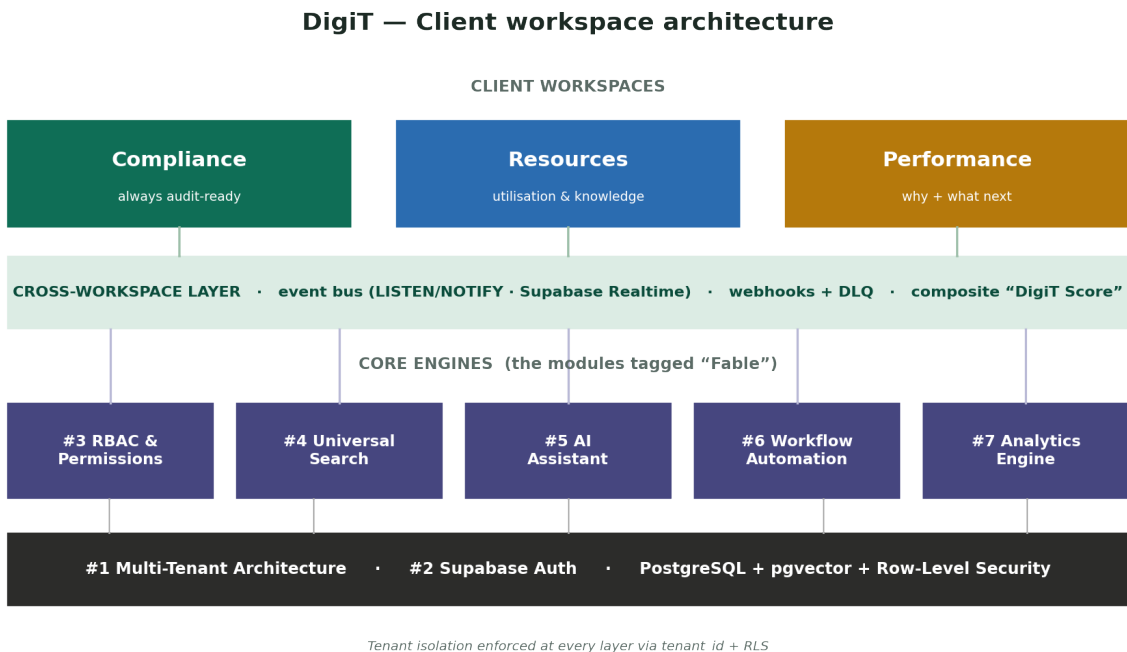


Figure 1 — Three workspaces, joined by a cross-workspace event layer, powered by five shared engines over a tenant-isolated Postgres foundation.

Compliance

Goal: continuous, self-assembling audit-readiness rather than pre-inspection scramble.

Data & security layer

- **Row-Level Security (RLS) policies** in Postgres keyed on `tenant_id` — the hard tenant-isolation boundary on every compliance record.
- **Append-only / immutable audit tables** with **hash-chained entries** (each row stores the SHA-256 of the previous) for tamper-evident trails — the 21 CFR Part 11 / GMP expectation.
- **Document versioning with content checksums + e-signature capture** (signed payload, signer ID, timestamp, reason-for-signing).
- **Object storage with signed URLs, TTL retention policies and legal-hold flags** for the evidence vault.
- **DSAR pipelines** (data export + cascade-delete) and **versioned consent records** for DPDP / GDPR.
- **ABAC (attribute-based access control)** layered on RBAC for document-level sensitivity.

AI / ML layer

- **NER + zero-shot classification** over ingested regulatory feeds to tag which rules apply to a given tenant.
- **Requirement traceability matrix** — an LLM maps each control in a framework (GMP / ISO / DPDP) to the tenant's stored evidence; gaps become Compliance-Score deductions.
- **Retrieval over a controls corpus** to power the mock-audit question/answer flow.

Automation layer

- **CAPA / deviation tracking as a finite-state machine** (Open → Investigation → Action → Verification → Closed) with **SLA timers**.
- **Weighted rule-based scoring engine** via **materialized views** for the live Compliance Health Score.
- **Scheduled compliance checks** via **pg_cron / Vercel Cron**, and **date-diff expiry triggers** feeding a notification queue (licence / certificate renewals).

Resources

Goal: a system that knows what is idle, at risk, or wasted — not a static inventory.

Data & security layer

- **Hierarchical asset registry** (adjacency-list or 1-tree taxonomy) with **QR / barcode / RFID asset tags**.
- **Inventory model with reorder points (ROP), EOQ and low-stock webhooks**.
- **OCR pipeline** for scanned documents → indexed metadata.

AI / ML layer

- **RAG over the tenant's document corpus**: chunking → **embeddings in pgvector** → **hybrid search (vector + tsvector keyword)** with a **reranking step** — the knowledge librarian.
- **Predictive maintenance: time-series anomaly detection** on usage/sensor telemetry (z-score / IQR → Isolation Forest / Prophet) with **RUL (Remaining Useful Life) estimation**; **MQTT / webhook** ingest for IoT-heavy clients (turbines, trucks, HPLCs).
- **Clause / key-value extraction** for contract intelligence (renewal dates, obligations, penalty clauses) → triggers reminders.

Automation layer

- **Capacity / utilisation metrics** (allocation %, capacity-vs-demand) as **time-series rollups**; optional **resource-levelling / constraint-based allocation** for rebalance suggestions.
- **Vendor scorecards** via **normalised weighted-KPI aggregation**.
- **Onboarding orchestration** — a workflow that provisions access, training and checklists via API calls.
- **Reservation / booking system** with **conflict detection** and **ICS / CalDAV sync**.
- **Skill matrix + gap analysis** (competency-vs-requirement matrix queries).

Performance

Goal: explain why a number moved and what to do next — beyond static dashboards.

Data & analytics layer

- **Semantic / metrics layer** — central metric definitions, calculated fields and **materialized views / OLAP-style rollups** so every chart reads from one source of truth.
- **ELT pipeline with incremental refresh and CDC (change data capture)** from operational tables into an analytics schema.
- **Event instrumentation** (event schema, funnels, retention curves) for product / usage analytics.

AI / ML layer

- **Text-to-SQL** over the semantic layer for ask-your-data (NL question → guarded SQL → chart).
- **Anomaly detection** on metric streams: **STL decomposition + z-score / IQR**, or Prophet / Isolation Forest, for proactive alerts.
- **Time-series forecasting** (ARIMA / Prophet / exponential smoothing) with **confidence intervals**; **Monte Carlo / parameterised what-if simulation** for scenario modelling.
- **Driver decomposition + correlation analysis** for root-cause, including **cross-workspace joins** (a Performance dip ↔ a Resource shortage ↔ a Compliance lapse).
- **LLM summarisation** for the auto-generated executive briefing.

Automation layer

- **Threshold + anomaly-based alerting** with **notification routing and escalation policies**.
- **Scheduled report generation** (Cron-triggered → email / Slack) with templated narratives.
- **Cohort-based benchmarking with k-anonymity / differential privacy** for safe cross-tenant comparisons.
- **OKR / KPI target tracking** with **pacing and variance analysis** under a **driver-tree** model.

Cross-workspace layer

What makes DigiT operational intelligence rather than three modules: the workspaces emit and react to each other's events.

- **Event bus / pub-sub** — Postgres **LISTEN / NOTIFY** or **Supabase Realtime** (or Redis Streams): each workspace **emits domain events** and others **subscribe and react** (a Compliance gap auto-creates a Resource task that surfaces as a Performance risk).
- **Outbound + inbound webhooks** with **idempotency keys, exponential-backoff retries and a dead-letter queue (DLQ)**.
- **Composite scoring service** computing the single “**DigiT Score**” as a weighted blend across the three workspaces.
- **Correlation IDs + structured logging** so one operation is traceable across all three workspaces.

Build matrix — which engine powers what

You are not building twenty new things; you are surfacing five engines through three workspace lenses.

Component group	Powered by module	Appears in
RLS / ABAC, tenant isolation, e-signatures, audit access	#3 Dynamic RBAC & Permissions	Compliance (+ all)
RAG, hybrid search, knowledge librarian, evidence retrieval	#4 Universal Search	Resources, Compliance
Text-to-SQL, mock-audit Q&A, clause extraction, summarisation	#5 AI Assistant	All three
CAPA FSM, onboarding orchestration, alerting, scheduled jobs, event reactions	#6 Workflow Automation	All three
Semantic/metrics layer, rollups, anomaly detection, forecasting, benchmarking	#7 Analytics Engine	Performance (+ Resources)
tenant_id, RLS foundation, auth, pgvector storage	#1 Multi-Tenant + #2 Supabase Auth	Foundation for all

Sequencing: build the five engines first (they are your #3–#7 priorities), then expose each through the relevant workspace. The cross-workspace event layer is what differentiates DigiT from single-point competitors — prioritise it once two workspaces are live.